RESEARCH PORTFOLIO

Farhin Chowdhury, DBA

Consumer Behavior & Mixed Method Research

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OVERVIEW

Selected Work

Researcher specializing in identifying user pain points, translating qualitative and behavioral data into actionable insights, and driving measurable improvements in engagement, conversion, and system performance.

Six case studies spanning qualitative depth, quantitative and mixed-method research, continuous product research, and independent product development.

01 Understanding Trust & Perception in Consumer Behavior

Doctoral Dissertation · Phenomenological study · 11 in-depth interviews · ATLAS.ti thematic analysis

02 Improving B2B Engagement Through Research & Experimentation

Cargo Chief · Mixed methods · 10+ A/B tests · Affinity mapping · Behavioral analytics

03 Turning High-Volume User Issues into Actionable Product Insights

Uber Technologies · Continuous research · Root cause analysis · Cross-functional synthesis

04 QuietShelf: An AI-Powered Reading Companion

Independent product build · Applying doctoral research on the say-do gap to personal recommendation design · Claude API · Supabase

05 Improving Logistics & Marketing Strategy for a National Retailer

MBA Capstone Project · Mixed methods · Primary survey (n=124) · Financial modeling · Team-based strategy consulting

06 Why Small Retail Shops Are Losing Customers to Big Retailers

Undergraduate Dissertation · Quantitative study · SPSS & EViews · Correlation and multiple regression analysis

CORE COMPETENCIES

Where Each Skill Shows Up

A cross-section of the six case studies against the research and product competencies they draw on.

Competency	01 Doctoral	02 Cargo Chief	03 Uber	04 QuietShelf	05 MBA Capstone	06 Undergrad
Stakeholder Management		●	●		●	
Psychology Background	●				●	●
Strategy & Roadmap		●	●	●	●	
Storytelling	●	●	●	●	●	●
Usability Testing		●	●	●		
Interviews	●					

● demonstrated directly in the case study · ● present as reflection, a named next step, or a secondary input rather than the primary method

CASE STUDY 01 · DOCTORAL RESEARCH

Understanding Trust & Perception in Consumer Behavior

Doctoral Dissertation

INTERVIEWS | PSYCHOLOGY BACKGROUND | STORYTELLING

> **Key finding:** Users expressed strong ethical values but consistently prioritized price and style at the point of purchase, revealing a structural gap between attitude and behavior that no amount of messaging alone can close.

Overview

The U.S. fashion industry, valued at over $1.7 trillion, faces increasing scrutiny for unethical practices. This qualitative phenomenological study explored how consumers perceive Corporate Social Responsibility (CSR) initiatives and whether those perceptions influence purchasing behavior, brand loyalty, and advocacy.

The Challenge

While fashion brands are investing billions in CSR, from ethical labor to environmental sustainability, there is a profound asymmetry of information. Brands invest heavily, yet consumers struggle to find, trust, or act on CSR information during the shopping experience.

Problem statement: Despite industry investment, consumer awareness of CSR efforts remains low, and brands struggle to translate ethical initiatives into measurable loyalty or advocacy.

Methodology

- **Approach:** Phenomenological design to uncover the lived experiences and mental models of fashion consumers
- **Participants:** 11 diverse U.S.-based consumers, recruited via purposeful and snowball sampling
- **Data collection:** In-depth, semi-structured interviews conducted via Zoom
- **Analysis:** Six-step thematic analysis using ATLAS.ti, synthesizing 100+ pages of transcripts into actionable themes
- **Rigor:** Member checks and a 32-item COREQ checklist to ensure transparency and trustworthiness

Key Insights

1. The Authenticity Barrier

Participants showed high awareness of CSR but deep skepticism toward brand claims, often viewing them as marketing facades. Authenticity was judged by third-party certification and consistent long-term behavior, not campaigns.

> *"I think that's kind of impossible to determine because at the end of the day, it's probably all performative."*
>
> Participant 4, on evaluating brand CSR authenticity

This captures the dominant user mental model: in the absence of verifiable proof, skepticism becomes the default. Designing for trust requires structural transparency, not better messaging.

2. The Priority Gap

Despite high ethical expectations, such as ending child labor and reducing pollution, CSR consistently fell behind price and style at the actual moment of purchase. Even ethically conscious participants made trade-offs when cost was involved.

> *"It would be a win if I could fulfill the requirement of a daily tee shirt from a less favorable CSR company and save $30 … spending $60 to $70 on a plain shirt is meaningless."*
>
> Participant 2, on price vs. ethics at the point of purchase

The say-do gap in action: a participant who genuinely values ethical consumption still chose price when it mattered. The barrier isn't awareness or values; it's accessibility.

3. Proof Beats Promises

Brands that embedded ethical proof into the product experience itself, such as Patagonia's repair tags and TOMS' one-for-one model visible at checkout, generated more genuine loyalty than those relying on campaigns or landing pages.

Impact

This study directly demonstrated what UX researchers encounter constantly: a significant gap between what users say they value and what drives their actual decisions. All 11 participants expressed genuine concern for ethical and environmental issues, yet most did not act on those values at the point of purchase.

The findings carry direct implications for designing for trust. Consumers don't evaluate authenticity through claims alone; they look for third-party verification, behavioral consistency, and visible proof over time. This suggests no amount of messaging optimization fixes a trust problem rooted in structural skepticism, a pattern with clear implications for transparency features, certification displays, and in-experience communication.

The study also surfaced a design opportunity: consumers who wanted to make ethical choices often couldn't, because the information was inaccessible or unverifiable at the moment of decision. The gap wasn't purely attitudinal; it was informational and experiential. These findings contribute to broader literature on the gap

between prosocial attitudes and behavior, a pattern that recurs across industries wherever users must evaluate trust before deciding.

Source: Chowdhury, F. S. (2024). *Fashion with Conscience: An Exploratory Study on Consumers' Perception of Corporate Social Responsibility in the Fashion Industry of the United States.* Doctoral Dissertation, Westcliff University.

CASE STUDY 02 · MIXED-METHOD RESEARCH

Improving B2B Engagement Through Research & Experimentation

Cargo Chief

STAKEHOLDER MANAGEMENT | STRATEGY & ROADMAP | STORYTELLING | *USABILITY TESTING (REFLECTION)*

- **Role:** Data Support Specialist (User Insights & Experimentation)
- **Scope:** B2B logistics web platform and email experiences
- **Feedback channels:** Product demos, support tickets, ongoing email/chat interactions
- **Synthesis:** Affinity mapping, thematic tagging, spreadsheet-based pattern tracking
- **Experiments:** 10+ A/B tests (subject lines, body copy, headlines, feature descriptions, layout)

+30%	+10%	3 → 8	↑
ENGAGEMENT	CONVERSION	NPS	EMAIL CTR

The Problem

Cargo Chief's B2B logistics platform had steady traffic but underperforming engagement and conversion. Client satisfaction was low and support volume was climbing. The same four friction points kept surfacing across demos, tickets, and email: how to search and filter loads, how carrier and broker connections worked, how pricing and quoting was structured, and how to track shipment status. These weren't isolated complaints; they signaled a systemic gap between how the product was presented and how users needed to understand it.

Research Approach

1. Multi-Channel Qualitative Feedback Collection

Feedback was gathered continuously across three channels: demos surfaced real-time confusion and hesitation; support tickets created a structured record of recurring problems; and email/chat revealed how clients described the product in their own words, often very differently from internal framing. Running all three simultaneously created a richer picture than any single source.

2. Synthesis: From Raw Feedback to Testable Themes

Notes were logged after each session, written feedback was tagged with informal labels, tagged items were grouped via affinity mapping, and high-frequency themes were ranked by volume and severity. Four friction themes emerged consistently:

- Value proposition gap: users couldn't quickly grasp how the platform solved their specific problems
- Feature language mismatch: internal product language didn't match how clients described their workflows
- Pricing opacity: frequent pricing questions signaled a discoverability failure, not a pricing one
- Email irrelevance: feature-led messaging reduced perceived relevance

3. From Themes to Hypotheses to a Test Roadmap

Each theme became a specific, testable hypothesis before any experiment was designed, ensuring tests were grounded in observed behavior, not internal assumptions. The four themes were sequenced into a prioritized experimentation roadmap:

- **Theme 1 → Hypothesis:** Leading website headlines with client outcomes rather than features will increase time on page and click-through. → *Test: value proposition & headline A/B test*
- **Theme 2 → Hypothesis:** Feature descriptions using clients' own language ("find available loads fast" vs. "advanced shipment filtering") will improve comprehension and engagement. → *Test: feature description language A/B test*
- **Theme 3 → Hypothesis:** Emails leading with a client problem rather than a product update will raise open and click rates. → *Test: problem-framed vs. feature-framed email A/B tests*

4. Behavioral Analysis

Amplitude dashboards tracked session depth, feature engagement, email CTR, and conversion week over week. Weekly reports connected research findings to product and marketing decisions in real time, keeping stakeholders across both teams aligned on what the data meant and what to test next.

Key Insights

1. Users couldn't see themselves in the product

The product did what it claimed, but the way it was described didn't match how users thought about their problems. A communication and framing gap, not a feature gap.

2. Problem-framed messaging consistently won

Content that led with a client pain point outperformed feature announcements across both email and web. Users responded to recognition of their problem before the solution.

3. Discoverability was a trust problem

Clients who couldn't find features assumed they didn't exist, even when they did. Low discoverability drove churn and escalations unrelated to actual product gaps.

4. Client language was the most reliable input

The highest-performing variants used exact phrases clients had used, not internally generated copy. Listening systematically is itself a form of research.

Impact & Reflection

The process produced a 30% increase in engagement, a 10% lift in conversion, and an NPS jump from 3 to 8, all without changes to the product itself. The gap between Cargo Chief's capabilities and user understanding was a communication problem that research-informed experimentation solved directly.

If running this again, I would formalize the synthesis structure from the start and introduce structured usability testing before the first A/B test. Usability testing and experimentation work best when the qualitative informs the quantitative, not the other way around.

CASE STUDY 03 · CONTINUOUS PRODUCT RESEARCH

Turning High-Volume User Issues into Actionable Product Insights

Uber Technologies Inc.

STAKEHOLDER MANAGEMENT | STORYTELLING | *STRATEGY & ROADMAP (PRIORITIZATION)* | *USABILITY TESTING (REFLECTION)*

- **Role:** User Support Specialist (User Insights Focus)
- **Methods:** Live issue resolution, root cause analysis, log/error analysis, pattern identification, cross-functional collaboration
- **Focus:** Translating recurring user pain points into product and system improvements: continuous, real-time user research

−90%	−45%	5–10 min	↑
SYSTEM ISSUES	ESCALATIONS	RESOLUTION	SATISFACTION

The Problem

Users interacting with customer-facing tools, including the customer portal, in-vehicle tablets, IVR systems, and routing tools, frequently experienced disruptions. Individual issues were resolved quickly, but there was no structured approach to identifying recurring patterns or distinguishing genuine bugs from usability problems. This led to repeated failures, rising support volume, and fixes that addressed symptoms rather than root causes.

Research Approach

The work combined continuous qualitative insight collection (capturing not just what failed but how users understood the failure), behavioral and system analysis (logs, error messages, usage patterns), root cause analysis across 100+ recurring issues, and stakeholder management across teams, communicating synthesized patterns, not isolated tickets, as a narrative product and engineering could act on.

Worked Example: Customer Portal Navigation

One representative issue, traced from raw signal to outcome:

Signal: A steady stream of tickets reported the portal was "broken" or that users "couldn't find" the status information they needed. On the surface, these looked like isolated user errors or scattered technical complaints.

Synthesis: Rather than treating each ticket individually, I tracked them as a group. Four signals converged: the same confusion across many users; users describing the same wrong path; tickets clustering around one area of the portal; and, most tellingly, users doing nothing technically wrong. The system worked as built; the layout was misleading them.

Diagnosis: This was a navigation and information-architecture problem, not a bug and not user error. The portal's structure didn't match users' mental model of where status information should live. The "technical" framing in tickets was masking a usability issue.

Action: Because a full navigation redesign sat outside the immediate timeline, I created a clear user-facing help guide and FAQ addressing the specific confusion points the analysis surfaced, directly reducing repeat tickets on that flow.

Outcome: Repeat tickets on the affected flow dropped, and the synthesized pattern was shared with product as evidence that the navigation itself, not the users, needed attention, positioning a permanent fix as a prioritized improvement.

This reflects the broader approach behind the 90% reduction in system issues: consistently moving from individual incidents to synthesized patterns, and from "users are struggling" to a specific, evidence-backed diagnosis.

Key Insights

1. Recurring issues were systemic, not isolated

Many "one-off" incidents were rooted in deeper design flaws. Treating them as a group revealed the pattern.

2. Many "technical" errors were usability issues

A significant share of reported bugs were driven by confusing workflows. Distinguishing genuine faults from design flaws was often the most valuable step.

3. User pain signaled product gaps before metrics did

Repeated complaints surfaced friction invisible in system metrics alone, an early-warning system for product issues.

Impact & Reflection

The approach drove a 90% reduction in system-related issues, a 45% drop in escalations, and a shift from reactive resolution toward proactive, pattern-driven improvement.

The portal example highlights a tradeoff I would handle differently with more authority: a help guide was right within my scope, but it treated the symptom while the navigation design remained the root cause. With a dedicated research function, the stronger path would have been to validate the problem through a quick usability test and bring that evidence to product, closing the loop from insight all the way to design change.

CASE STUDY 04 · INDEPENDENT PRODUCT BUILD

NEW PROJECT

QuietShelf

A quieter way to remember what you've read.

AI-Powered Book Recommendation App · Status: In Development · Started 2026 · Builder: Farhin Chowdhury, DBA

STRATEGY & ROADMAP | STORYTELLING | *USABILITY TESTING (PLANNED)*

The Problem

I found my love for reading in 2022. What I didn't expect was the feeling that comes after finishing a book that truly moved me: a kind of emptiness, like something had ended that I wasn't ready to let go of.

Some books pull you into a completely different world. When they end, you don't just want another book; you want that specific feeling again. The problem is, no existing platform knows how to find it for you. Amazon recommends what you bought. Goodreads recommends what's popular. Neither of them understands what made a book matter to you in the first place.

That's the problem QuietShelf is built to solve.

The Research Behind It

My doctoral research in consumer behavior taught me that there is almost always a gap between what people say they want and what actually resonates with them. Preferences are contextual, emotional, and deeply personal, and they shift over time.

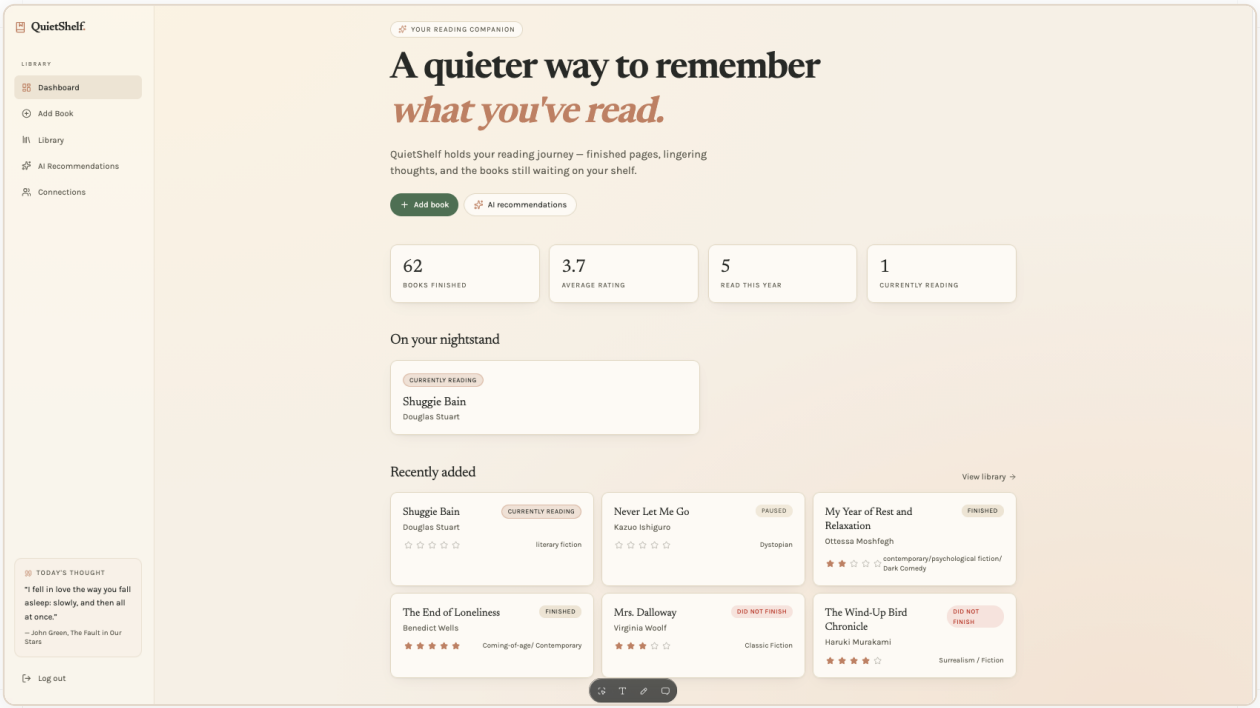
Most recommendation engines treat reading as a transactional behavior: you bought X, so you might like Y. But reading, at its best, is an emotional experience. The books that stay with you aren't always the ones that fit neatly into a genre; they're the ones that matched something you were feeling, thinking, or working through at a particular moment.

QuietShelf is built on that insight. Instead of optimizing for what's trending or what's frequently purchased together, it aims to understand how you engage with books over time and surface recommendations that fit the reader, not just the genre.

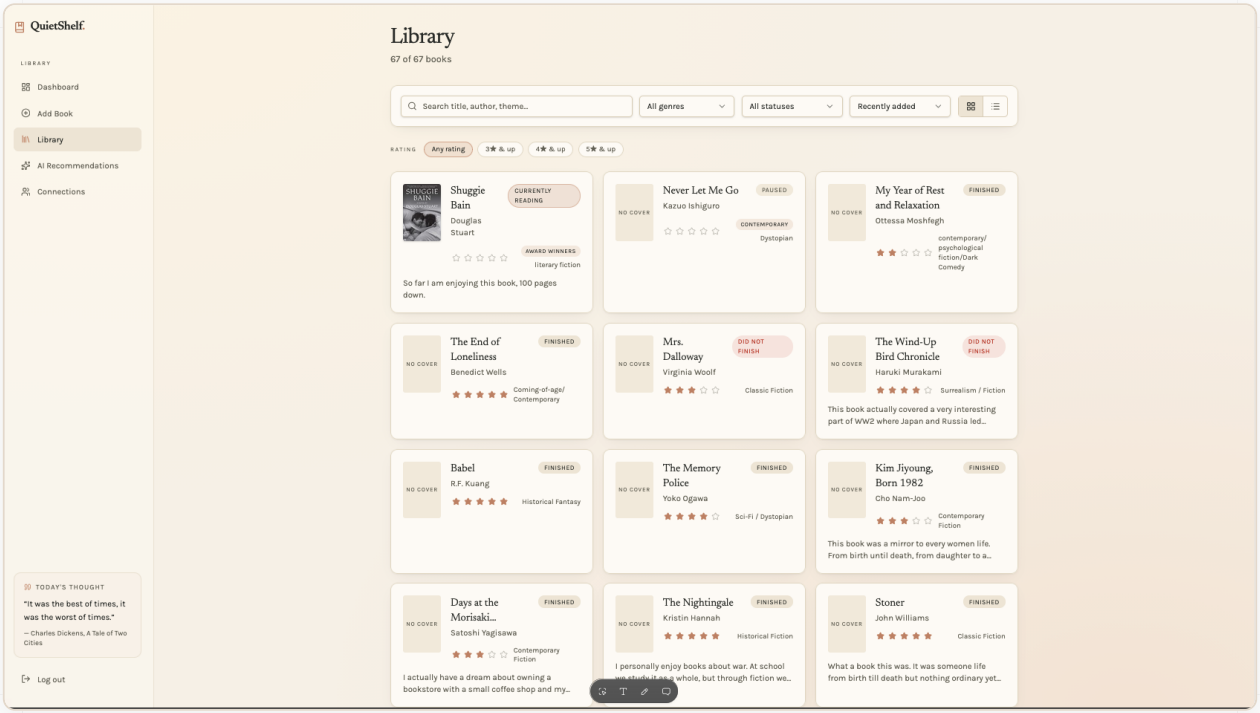
Product Overview

QuietShelf is a personal reading companion that learns your reading style and delivers book recommendations based on your individual patterns, not popularity, not purchase history. The core idea: the more you use it, the better it understands what kinds of books ignite something in you, and the better it gets at finding your next one.

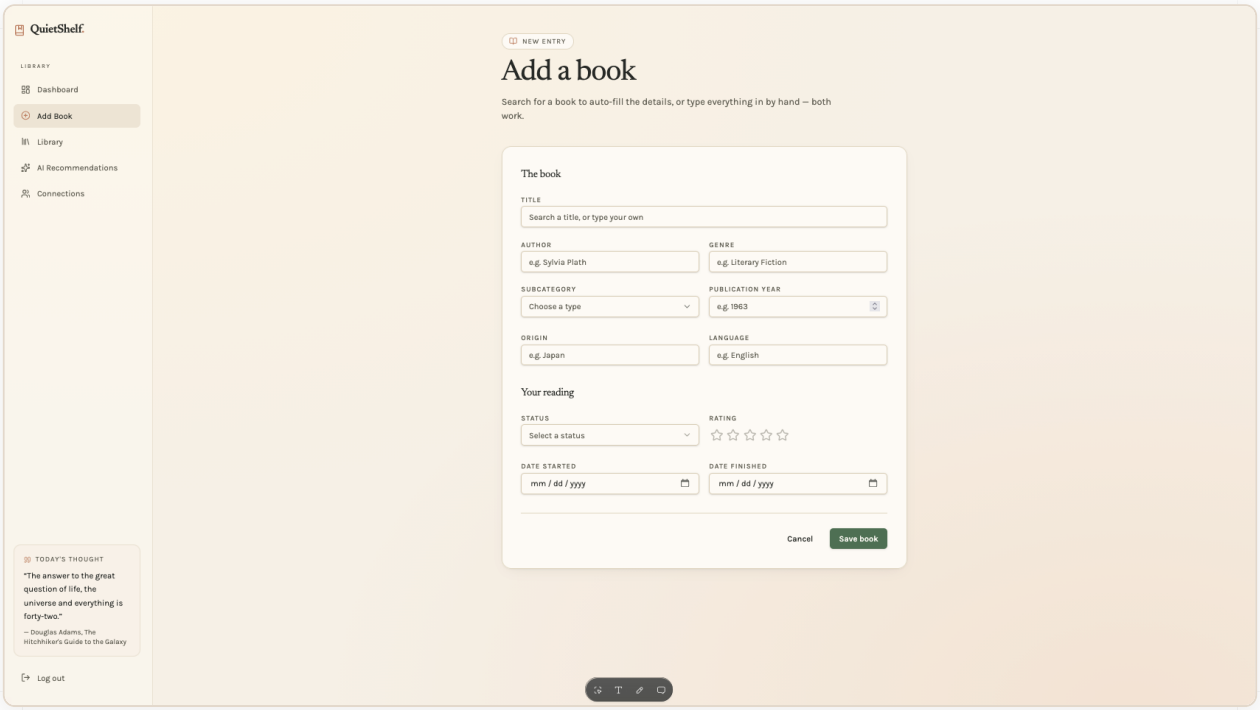
Product Walkthrough



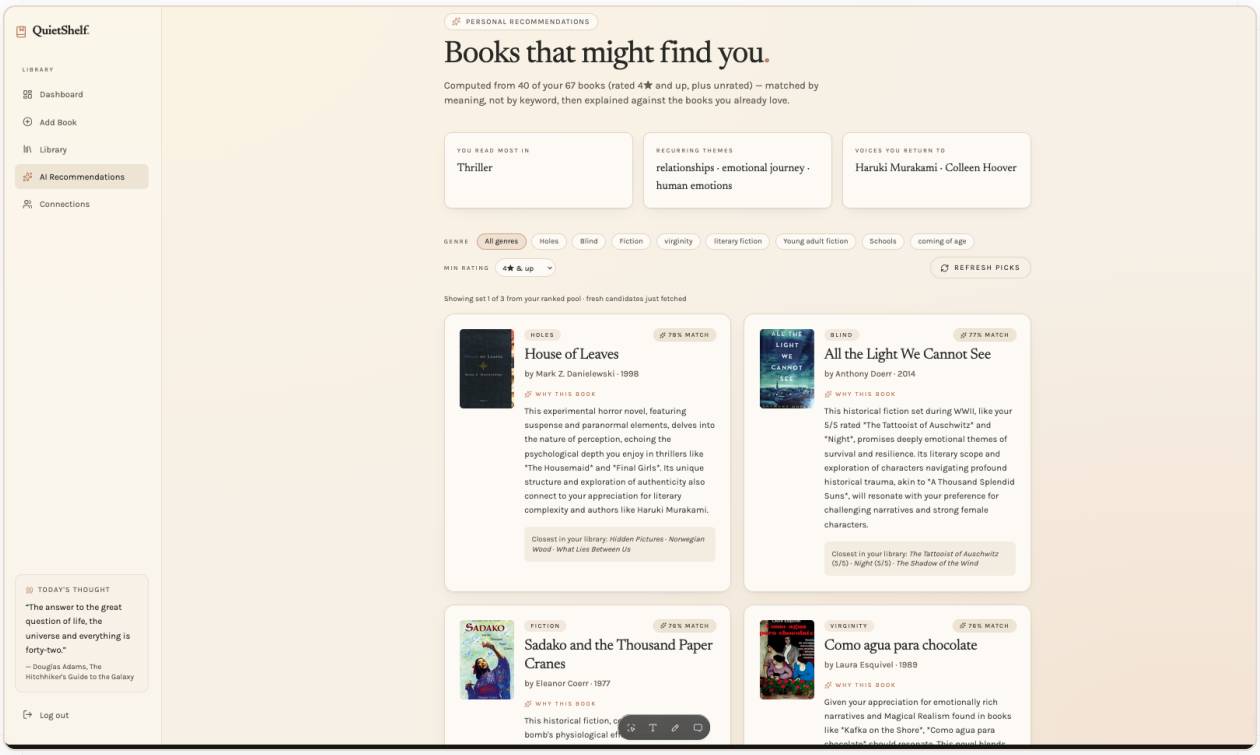
Dashboard: tracks finished books, ratings, and reading stats at a glance, including 62 books finished, a 3.7 average rating, and what's currently on the nightstand.



Library: a full reading log of 67 books with ratings, genres, and status (finished, currently reading, paused, did not finish), filterable and searchable.



Add Book: a simple entry flow with search-to-autofill or manual entry, capturing genre, subcategory, origin, language, reading status, rating, and dates.



AI Recommendations: personalized matches computed from the reader's rated library, with a stated match percentage and a "why this book" explanation grounded in specific titles already in the library.

Build Journey

Year	Milestone
2022	Started keeping a personal reading log; noticed the gap in existing recommendation tools.
Early 2026	Began designing QuietShelf; built the first prototype using Lovable to validate the concept quickly.
Now	Rebuilding from scratch using the Claude API for AI-driven recommendations and Supabase as the backend, moving from proof of concept to a fully functional product.

What's Next

- Launch a working version with real AI recommendation logic powered by the Claude API
- Test with a small group of readers to gather behavioral data and refine the recommendation model
- Explore how reading mood and context (time of year, emotional state, pace) influence what feels like the right book
- Build a community feature for readers who want to share what a book meant to them, not just rate it

Skills & Tools

Area	Details
Research	Consumer behavior analysis, user-centered design, iterative prototyping
Tools	Lovable, Claude API, Supabase
Methods	Qualitative insight, behavioral pattern analysis, product research

Built by Farhin Chowdhury, DBA, Consumer Behavior Researcher & UX Researcher
Portfolio: farhin.notion.site

CASE STUDY 05 · MBA CAPSTONE PROJECT

Improving Logistics & Marketing Strategy for a National Retailer

Cameron University MBA Capstone · Team-Based Strategy Consulting Engagement

STRATEGY & ROADMAP | STAKEHOLDER MANAGEMENT | STORYTELLING

PSYCHOLOGY BACKGROUND (PERSONA, PAIN POINTS)

> **Key finding:** Tuesday Morning's profitability problem was not a demand problem. A fragmented distribution network and an under-leveraged loyalty and social presence, not weak pricing or product, were quietly eroding margin and brand awareness.

Overview

As the capstone project for a nine-person MBA consulting team (operating under the name SOL Consulting), this engagement delivered a strategic turnaround plan for Tuesday Morning, a national off-price "treasure hunt" retailer. The team combined industry and competitive analysis with primary survey research and financial modeling to propose two flagship recommendations: a new distribution center to fix a structural logistics gap, and a repositioned marketing strategy grounded in a validated customer profile.

Methodology

- **Approach:** Mixed-methods strategy consulting, combining qualitative industry and competitive analysis (Porter's Five Forces, PESTLE, SWOT) with quantitative financial and logistics modeling
- **Primary research:** Self-administered survey of 124 respondents, recruited via Facebook and word-of-mouth, used to validate the target customer profile
- **Secondary research:** Competitor benchmarking against Ross, Big Lots, and TJ Maxx; financial statement analysis; social media and brand-presence audit
- **Modeling:** A five-year phased cost-savings model for a proposed distribution center, and a five-year marketing budget reallocation model

Key Insights

1. The logistics network was a hidden margin problem

Two distribution centers were serving stores across the entire country, driving up cost of goods sold through excess Store-to-Distribution-Center mileage. Modeling a phased-in third distribution center in Atlanta, Georgia projected roughly $24 million a year in transportation savings once fully integrated, alongside a five-year workforce transition plan between the Dallas and Atlanta facilities.

2. The survey validated (and sharpened) the target customer

Rather than assume who the "Bargain Hunter" customer was, the team tested it: the 124-respondent survey reinforced a target profile of women aged 35 to 65, upper-middle income, motivated less by price alone than by the thrill of the treasure hunt itself. That distinction, experience over pure discount, reshaped the marketing brief.

3. Awareness, not spend, was the real marketing gap

Tuesday Morning was already spending about 2% more of sales on advertising than competitors, yet had a fraction of their social following (roughly 200,000 Facebook likes versus a competitor's 2.5 million). The problem wasn't marketing investment; it was where that investment was going. The team's plan reallocated budget toward loyalty programs, social media, and targeted print rather than simply spending more.

Impact & Reflection

The final proposal paired a quantified operations fix (the distribution center model) with a research-validated marketing strategy (the survey-informed customer profile and reallocated budget), giving the client a plan grounded in evidence rather than intuition on both the cost and revenue sides of the business.

This project was an early exercise in the same synthesis skill that carries through the rest of this portfolio: taking fragmented, anecdotal signals (rent expense, inventory turnover, low social engagement) and turning them into one coherent, quantified strategic narrative.

CASE STUDY 06 · UNDERGRADUATE DISSERTATION · QUANTITATIVE RESEARCH

Why Small Retail Shops Are Losing Customers to Big Retailers

An Inferential Study in the Context of Nepal

PSYCHOLOGY BACKGROUND | *STORYTELLING (FINDINGS TO RECOMMENDATION)*

> **Key finding:** Product quality, not price, was the strongest predictor of why customers defected to big retail chains (beta = .678, p < .001), ahead of both service and price.

Overview

Small, independently owned retail shops in Nepal were steadily losing customers to large all-in-one chains such as Bhatbhateni, Bigmart, and Saleways. This quantitative dissertation set out to identify and rank the factors driving that switch, grounded in the Howard-Sheth Theory of Buyer Behavior, so that small retailers would know where to focus limited resources.

Methodology

- **Approach:** Quantitative, correlational research design testing three hypotheses about the drivers of big-retailer preference
- **Sample:** 71 customers surveyed via structured questionnaire across 10 small retail shops in Kathmandu, with 7 to 8 respondents sampled per shop
- **Model:** Customer preference for big retailers modeled as a function of three independent variables: service, price, and product quality
- **Analysis:** Descriptive statistics, correlation analysis, and multiple linear regression, processed in SPSS and EViews

Key Insights

1. Quality dominated the model

Of the three predictors tested, product quality had by far the strongest and most significant relationship with customer preference (beta = .678, p < .001). The overall model explained about 60% of the variance in satisfaction with big retailers (R squared = .599), and the ANOVA confirmed the model as a whole was highly significant (F = 33.379, p < .001).

2. Service mattered, but only marginally

Service showed a positive relationship with big-retailer preference, but it fell just short of conventional statistical significance ($p = .069$), suggesting it is a real but secondary lever rather than the primary driver.

3. Price was not the deciding factor

Price range was the weakest of the three predictors and did not reach significance ($p = .139$), directly challenging the common assumption that small shops lose customers mainly on price.

Impact & Reflection

The findings reframed the problem for small retailers: rather than compete on price against chains with far greater scale, a more promising strategy was to invest in the perceived quality of what they carry, with service improvements as a secondary priority. The study's own limitations (a modest sample of 71, self-reported responses, and a single city) are acknowledged directly in the dissertation as boundaries on how far the findings generalize.

This was an early, formative exercise in the same quantitative toolkit, SPSS, correlation, and regression, that later fed into the mixed-methods research approach used throughout this portfolio.